

PVsyst - Simulation report

Grid-Connected System

Project: PV_Hydrogen_Cyprus_10MWp

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 10000 kWp

Aglantziá - Cyprus



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PVsyst V8.1.5

VC1, Simulation date:
10/09/26 17:50
with V8.1.5

Project summary

Geographical Site	Situation	Project settings
Aglantziá	Latitude 35.1474 °(N)	Albedo 0.20
Cyprus	Longitude 33.4002 °(E)	
	Altitude 173 m	
	Time zone UTC+2	
Weather data		
Aglantziá		
PVGIS api TMY		

System summary

Grid-Connected System	No 3D scene defined, no shadings		
Orientation #1	Near Shadings	User's needs	
Fixed plane	no Shadings	Unlimited load (grid)	
Tilt/Azimuth 28 / 0 °			
System information			
PV Array		Inverters	
Nb. of modules 16666 units		Nb. of units 24 units	
Pnom total 10000 kWp		Total power 8448 kWac	
		Pnom ratio 1.18	

Results summary

Simulation step	Hourly				
Produced Energy	19174156 kWh/year	Specific production	1917 kWh/kWp/year	Perf. Ratio PR	87.31 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 28 / 0 °

Models used

Transposition

Diffuse

Circumsolar

Perez

Imported

separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer

Generic

Model

TSM-DE20-600

(Original PVsyst database)

Unit Nom. Power

600 Wp

Number of PV modules

16666 units

Nominal (STC)

10000 kWp

Modules

641 units x 26 In series

At operating cond. (50°C)

Pmpp

9146 kWp

U mpp

812 V

I mpp

11258 A

Total PV power

Nominal (STC)

10000 kWp

Total

16666 modules

Module area

47167 m²

Cell area

44098 m²

Inverter

Manufacturer

Generic

Model

SG350-HX

(Original PVsyst database)

Unit Nom. Power

352 kWac

Number of inverters

24 units

Total power

8448 kWac

Operating voltage

500-1450 V

Pnom ratio (DC:AC)

1.18

Power sharing within this inverter

Total inverter power

Total power

8448 kWac

Number of inverters

24 units

Pnom ratio

1.18

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)

29.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s

DC wiring losses

Global array res.

1.2 mΩ

Loss Fraction

1.50 % at STC

Module Quality Loss

Loss Fraction

-0.38 %

Module mismatch losses

Loss Fraction

2.00 % at MPP

Strings Mismatch loss

Loss Fraction

0.05 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

Simulation step

Hourly

System Production

Produced Energy

19174156 kWh/year

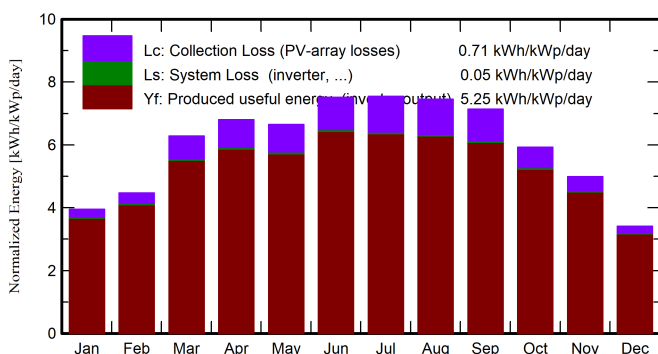
Specific production

1917 kWh/kWp/year

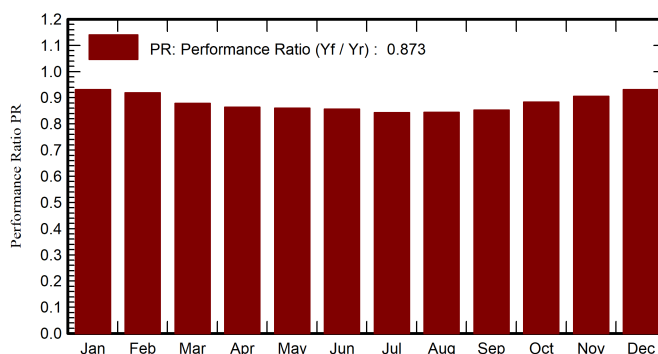
Perf. Ratio PR

87.31 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	82.5	35.63	10.32	122.4	120.5	1151610	1140072	0.931
February	95.0	44.88	11.47	125.1	122.8	1159998	1148436	0.918
March	162.5	56.07	14.68	194.9	191.3	1727034	1711418	0.878
April	191.0	67.58	19.58	204.0	200.0	1777665	1761058	0.863
May	211.4	73.45	23.85	206.2	201.6	1791799	1774078	0.860
June	241.4	65.87	25.81	225.4	220.2	1949578	1931288	0.857
July	244.6	64.78	31.12	233.9	229.0	1990021	1971345	0.843
August	223.2	61.01	29.74	231.1	226.4	1968007	1949830	0.844
September	184.0	53.09	27.11	214.2	210.0	1842211	1825753	0.853
October	139.1	44.92	21.49	183.7	181.0	1637065	1622012	0.883
November	100.7	37.19	17.54	149.6	147.1	1366341	1353643	0.905
December	70.5	35.83	11.26	105.8	103.7	996031	985222	0.931
Year	1945.9	640.30	20.38	2196.2	2153.6	19357358	19174156	0.873

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

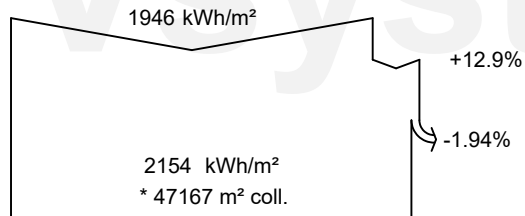
PR Performance Ratio



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Loss diagram



Global horizontal irradiation

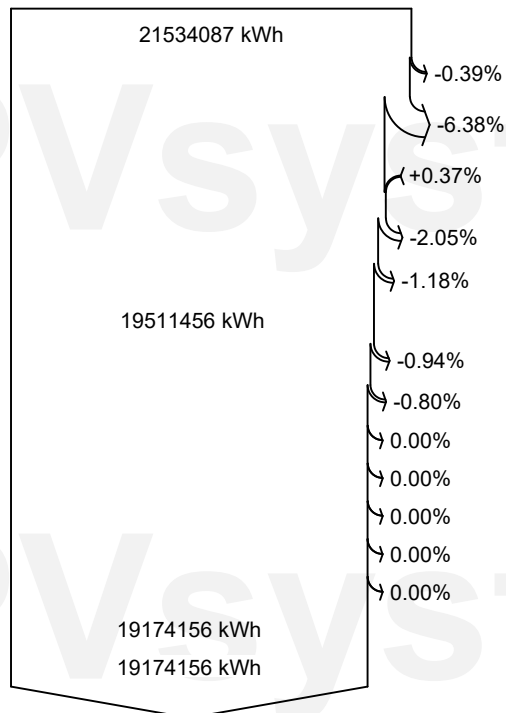
Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

efficiency at STC = 21.20%



Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

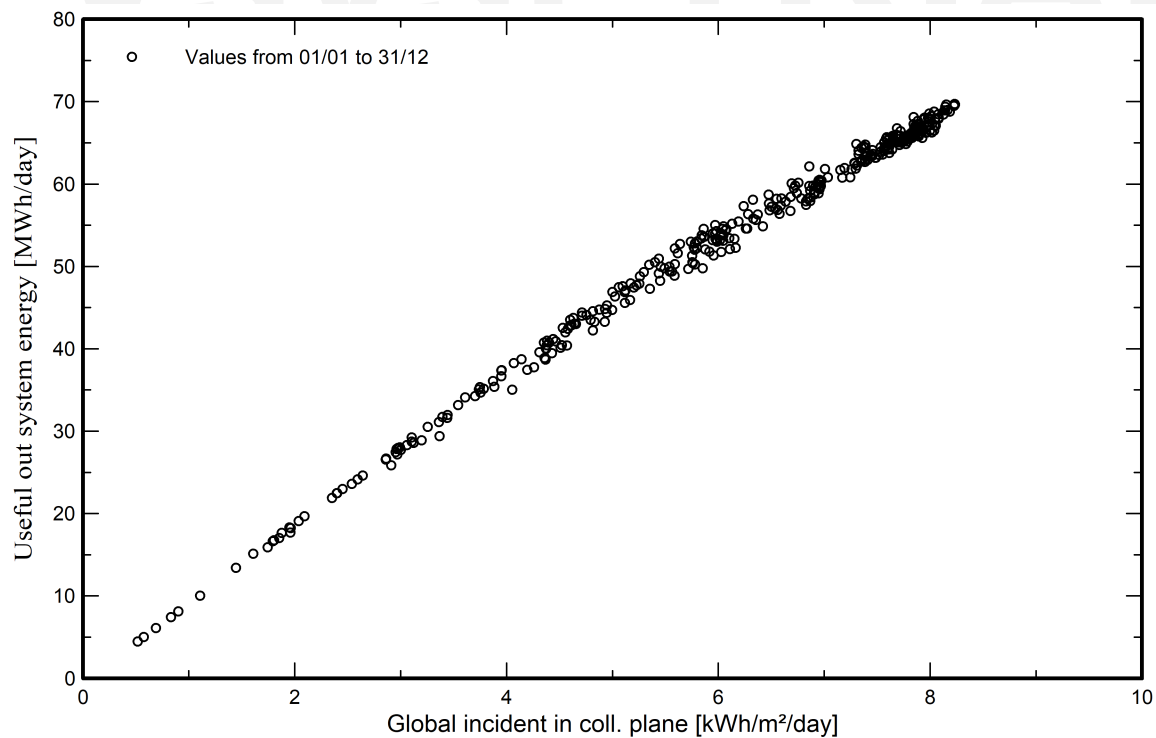
Energy injected into grid



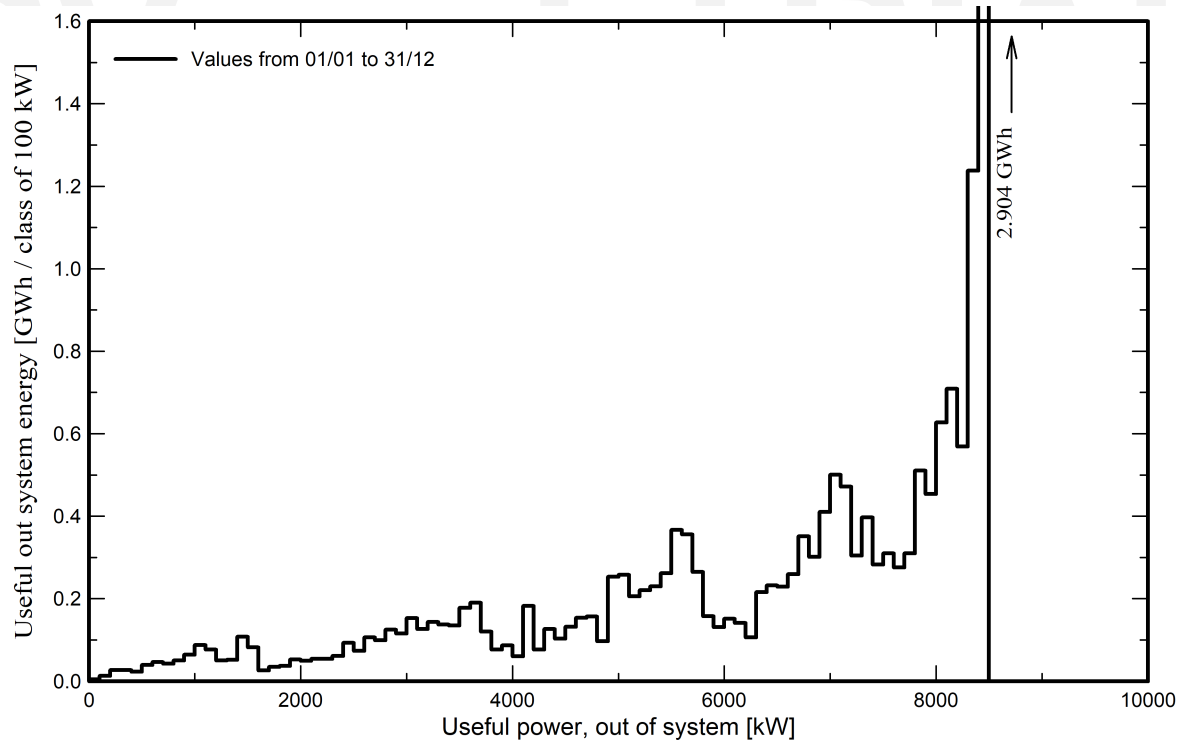
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

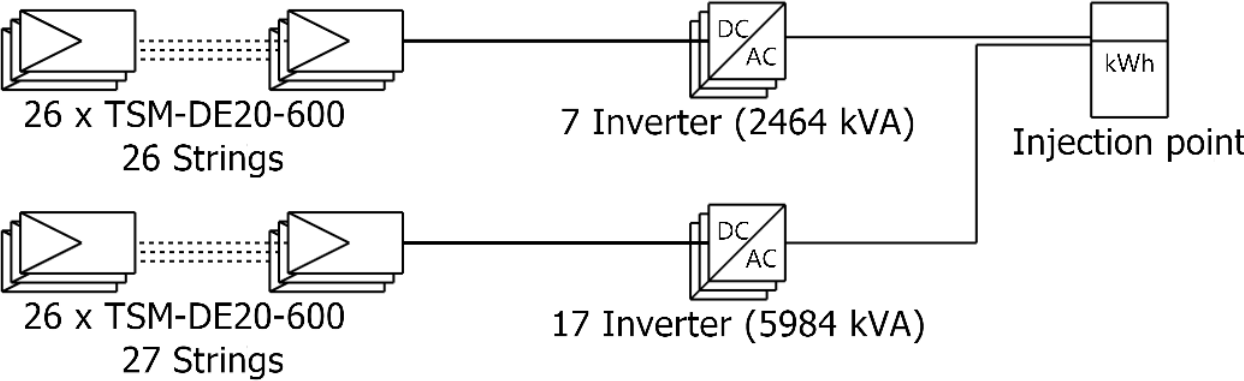




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Single-line diagram



PV module	TSM-DE20-600
Inverter	SG350-HX
String	26 x TSM-DE20-600

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